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# Fission-fusion and the evolution of hominin social systems

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ABSTRACT

The course of hominin evolution has involved successive migrations towards higher absolute latitudes over the past three million years. Poorer habitat quality further from the equator has led to the necessity for groups occupying higher latitudes to live at lower population densities. Coupled with a trend towards increasing group size over this time period, this tendency towards expansion has led to exponential increases in the area requirements of hominin groups, and a concomitant need to adjust foraging patterns. The current analyses suggest that the development of increasingly complex, multi-level fission-fusion social systems could have freed hominins of the foraging constraints imposed by large group sizes and low population densities. Analyses of the fossil record suggest latitudinally-driven differences in area requirements of the australopithecines from East and South Africa, and African and Asian *Homo erectus*. In contrast, chronologically-driven differences appear between *H. erectus* as a whole and *Homo heidelbergensis*, and between *H. heidelbergensis* and the Neanderthals. These results are discussed in relation to studies of the foraging patterns of primates and hunter-gatherers.

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## Introduction

The course of evolution has taken hominins out of Africa on at least two occasions, and hominin evolution generally has been a story of successive migrations towards higher absolute latitudes (Gamble, 1993, 2009; Lahr and Foley, 1994, 1998; Templeton, 2002). Decreasing diversity and density of species further from the equator are recurrent ecological patterns (Wallace, 1891; Dobzhansky, 1950; Pianka, 1966; Rapoport, 1982; Stevens, 1989; Willig et al., 2003), correlating with reductions in effective temperature (Bailey, 1960; Gregorczuk and Cena, 1967) and suggesting that hominins expanding out of the tropics would have been forced to live at lower population densities, ranging further to acquire necessary resources (Binford, 1980; Kelly, 1983, 1995; Kusimba, 2003). This trend towards latitudinal expansion has been parallelled by an exponential increase in hominin group sizes related to the increases in both absolute and relative brain size over the past three million years (Aiello and Dunbar, 1993; Dunbar, 1998, 2003; Grove, 2010a). The areas required to sustain hominin groups as they increased in size and moved towards higher latitudes over this period must therefore have expanded substantially, with concomitant effects on the length of time required to complete the foraging round. Studies of both social carnivores (Gittleman, 1989; Wrangham et al., 1993) and primates (Chapman, 1990; Chapman et al., 1995; Gillespie and Chapman, 2001) have demonstrated that increases in day journey length caused by group size increases are unsustainable and lead to group fission. Hunter-gatherers, however, sustain communities of sizes unparallelled among either of these groups (Murdock, 1949; Birdsell, 1958, 1968; Binford, 2001), maintaining links with neighbouring communities over vast areas (Whallon, 2006). An understanding of the mechanisms by which such communities are sustained is of vital importance to any scenario of human social evolution.

The seed of a potential solution to the problem of exponentially increasing area requirements during the course of human evolution can nevertheless be found in the social organization of precisely those species that are documented to fission under the pressures of increased group size. Primates such as chimpanzees (Nishida, 1968; Symington, 1990; Lehmann and Boesch, 2004), bonobos (Kano, 1982; Furuichi, 2009), and spider monkeys (Klein and Klein, 1977; Shimooka, 2003; Wallace, 2008), as well as carnivores such as coatis (Gompper, 1996), kinkajous (Kays and Gittleman, 2001), and hyenas (Kruuk, 1976; Mills, 1990; Smith et al., 2008), fission only temporarily into foraging subgroups whilst retaining longer-term membership of wider communities. Such 'fission-fusion' social organization allows these species to simultaneously reap the benefits of living in larger communities whilst avoiding many of the associated costs. Whilst benefits of larger groups could include the

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reduction of predation risk through enhanced vigilance (Pulliam, 1973), dilution of per capita risk (Foster and Treherne, 1981), or predator confusion (Krakauer, 1995), as well as access to a larger pool of potential mates (Wiley, 1991) and enhanced success in inter-group competition (Alexander, 1974), the costs of grouping relate almost exclusively to foraging (e.g., Free et al., 1977; Ruxton, 1995; Broom and Ruxton, 1998; Stillman et al., 2000; Lehmann et al., 2007; though see Templeton and Giraldeau, 1995, 1996).

The process of periodic splitting and re-aggregation of groups provides a means of ensuring that group size is optimal at any given time and for any given task, and is practiced by the vast majority of human hunting and gathering groups (Binford, 2001; Grove, 2009, 2010b; Layton and O'Hara, 2010). However, a visible difference between the fission-fusion systems of humans and other primates lies in the number of 'levels' into which hunter-gatherer societies are divided. Whilst primates, carnivores, and other fission-fusion mammals maintain only two levels – nominally the community and the foraging group, where the latter is a subgroup of the former – human hunter-gatherers operate using a minimum of four grouping levels. In one of the earliest descriptions of such a system, Murdock (1949) postulates a hierarchy of nuclear and composite families, consanguineal kin groups, clans, and communities, whilst Birdsell's (1958, 1968) seminal 'magic numbers' also suggest four levels of grouping. These various levels recur in ethnographic studies (e.g., Birdsell, 1953; Stanner, 1965; Tindale, 1974; Williams, 1974; Burch, 1975; Dixon, 1976), and are robust to various analytical methods (Zhou et al., 2005; Hamilton et al., 2007; Grove, 2010c, 2011a). An important aspect of the multiple levels of human grouping is that they are 'hierarchically inclusive' (Dunbar, 1998), forming nested series where, to take Murdock's (1949) example, a community is composed of a number of clans, which are in turn composed of a number of consanguineal kin groups, and so on through the levels.

If hunter-gatherer fission, like that of other mammals, is a solution to foraging pressures, how and when might these various levels have emerged? Layton and O'Hara (2010) suggest that the band emerged as an optimal solution to the conflict between dispersal and cooperation, where the former relates to the decreased population density and greater mobility necessitated by an increased reliance on hunting (e.g., Grove, 2009, 2010b) and the latter involves both a division of labour and the practice of reducing the variance in hunting returns by sharing meat (e.g., Kaplan and Hill, 1985; Kaplan et al., 1990). A community of increasing size becomes non-viable as a cohesive foraging unit when the area of habitat required to sustain it cannot be covered in some reasonable period of time. At this point, the community must fission, producing a number of subgroups that can more efficiently cover the required area. Layton (in press) suggests that "the critical point will occur when the daily foraging range is less than the area occupied by the community", as at this point foragers encounter a fundamental ecological constraint. A similar effect will occur when the density of resources is reduced, as this effectively increases inter-patch distances, requiring a group of a given size to travel further to gain sufficient energy. Decreasing primary resource density will tend to lead to decreasing density of the consumer, with the cascading effect that animals at yet higher trophic levels (i.e., secondary consumers) will be forced to live at yet lower population densities due to inefficiencies of energy assimilation (Elton, 1927). It is clear, then, that both increasing the size of a group and decreasing the density at which its constituent individuals can live will lead ultimately to increases in the area that the group requires for the energetic fulfilment of its members.

In modelling the combined effects of increasing group size and expansion towards higher latitudes on evolving hominin groups, the crucial variable therefore becomes the area required to sustain a group of a given size at a given population density. Since group size is expected to increase towards the present, and population density to decrease as groups move further from the equator, the overall expectation is of rapid increases in area requirements as human groups proceeded in spatio-temporal expansions from sub-Saharan Africa towards Eurasia. Though the relationship between group size and age is approximately exponential (Aiello and Dunbar, 1993), the shape of the relationship between population density and latitude among hunter-gatherers has not previously been studied in any detail. Having modelled latitudinal and temporal variations in areal requirements, conclusions concerning fission-fusion dynamics are calculated using gas model approximations to examine foraging efficiency when the overall community is split into various subgroup configurations. Finally, data on hominin fossils of known latitude and age are plotted against the fission thresholds thus established in order to provide predictions regarding this aspect of ancestral social systems.

## Methods

The first step in estimating the area required for a group of a given age living at a given latitude involves an expansion of Aiello and Dunbar's (1993) analysis of the relationship between age, endocranial capacity, brain size, neocortex ratio, and group size. A new dataset consisting of 191 endocranial capacities from dated fossil hominin crania was subjected to a revised version of the process outlined by Aiello and Dunbar (1993). This dataset contains a diverse array of hominin species from *Australopithecus afarensis* to *Homo sapiens*, spanning a period of approximately 3.2 Ma (millions of years). Regrettably, there is a lack of available data on the genus *Paranthropus*, and we hope to be able to remedy this problem in the future. Thankfully, however, this single weakness of the record does not prevent us from carrying out a detailed and valuable study. As a number of the equations employed by Aiello and Dunbar (1993) were published in older work (e.g., Martin, 1990), where they were accompanied neither by the appropriate confidence intervals nor the subsidiary statistics required for their calculation, data on primate brain volume (Stephan et al., 1981), endocranial capacity (Martin, 1990), neocortex ratio (Stephan et al., 1981; Dunbar, 1992a), and group size (Dunbar, 1992a, 2009) were re-assembled from the original literature and regression equations re-calculated. Firstly, brain volume ($bv$, $mm^3$) was regressed on endocranial capacity ($ec$, $mm^3$), yielding the following equation:

$$\log10(bv) = 0.9769\log10(ec) + 0.1073 \quad (1)$$

($n = 30$, $F_{1,28} = 9454.741$, $R^2 = 0.997$, $p < 0.001$). Neocortex ratio ($nr$) was then calculated from brain volume as:

$$\log10(nr) = 0.1748\log10(bv) - 0.4490 \quad (2)$$

($n = 24$, $F_{1,22} = 180.512$, $R^2 = 0.891$, $p < 0.001$). Finally, group size ($gs$) was calculated from neocortex ratio using additions from Dunbar's (2009) revised dataset, as:

$$\log10(gs) = 3.1953\log10(nr) + 0.1527 \quad (3)$$

($n = 27$, $F_{1,25} = 783.819$, $R^2 = 0.922$, $p < 0.001$). A potential problem with a series of regressions where each new result depends on the result of the last is the multiplication of the confidence intervals associated with the estimates (Smith, 1996). Furthermore, the individual confidence intervals for estimates that constitute new observations (i.e., observations not present in the data used to calculate the regression equation) can be substantial. In the present instance, the high coefficients of determination of Eqs. (1)–(3) mean that this problem is not limiting. Nonetheless,

full individual confidence intervals for each stage of the modelling are given in the results section below.

Eqs. (1)–(3) allow us to utilize the functional relationship established by Dunbar (1992a, 1993) between neocortex ratio and group size to predict group sizes of extinct hominins (as per Aiello and Dunbar, 1993). However, given that what we actually measure directly in the fossil record is endocranial capacity, it is useful also to provide an equation that allows researchers to calculate group size directly from endocranial capacity. This equation is given as:

$$\log10(gs) = -0.986 + 0.449\log10(ec) \quad (4)$$

($n = 27$, $F_{1,25} = 117.433$, $R^2 = 0.313$, $p < 0.004$). Although this equation is not employed in the analyses presented in this paper, graphical comparisons of results obtained via this equation are compared with those using the functionally correct route of Eqs. (1)–(3) in SOM.

Eqs. (1)–(3) provide a link between endocranial capacity and group size, and form the first element of the calculation of the area required per group. The second element involves an examination of the relationship between latitude and population density among modern hunting and gathering groups. For this second element, data from Binford's (2001) compendium were used. These data describe modern hunting and gathering peoples, and we should note that such peoples may not experience precisely the same constraints as our hominin ancestors in prehistory. Many of the ancestors comprising our fossil dataset were of different species, with different body sizes, technologies, and dietary strategies. The large size of the (Binford, 2001) sample and the variation in body size, technology, and dietary strategy within it suggest that constraints may have been functionally similar. However, the range of variation in the (Binford, 2001) sample is almost certainly smaller than that in the fossil dataset, and we must bear this in mind when interpreting results.

Of the 339 cases listed in Binford's sample, those that were not fully mobile (GRRPAT = 2 in Binford's Table 5.01, $n = 87$) were excluded, as were those that were listed as 'suspect cases' ($n = 54$) and those described as 'mounted hunters' ($n = 17$). These exclusion criteria yield a sample of 181 groups. The variation in population density is relatively high among groups near the equator, with this variation reducing substantially as absolute latitude increases. This pattern implies that the latitudinal constraint on population density is most clearly manifest among those groups living at high population densities for a given latitude, and suggests that the best regression model for the data is an upper bound equation (Blackburn et al., 1992). Simulation studies by Blackburn et al. (1992) suggest that datasets characterized by a marked upper bound can be accurately modelled by dividing the data into equal bins along the abscissa and calculating a regression equation using the maximum value within each bin. Though this procedure results in a drop in regression sample size, which could conceivably reduce both the coefficient of determination and the probability that the F-statistic is significant, the advantage is a substantially more meaningful model for the dataset. Blackburn et al. (1992) find that equal division into eight bins provides a suitable compromise between accuracy and precision. Their suggestion is followed here, resulting in a relationship between maximum population density ($pd$; individuals per km$^2$) and absolute latitude (abslat) described by the following equation:

$$pd = -0.0104\,\text{abslat} + 0.8636 \quad (5)$$

($n = 8$, $F_{1,6} = 11.416$, $R^2 = 0.655$, $p = 0.015$). This relationship, including the upper bound produced following the method of Blackburn et al. (1992), is shown in Fig. 1.

Given the relationships depicted in Eqs. (3) and (5), the area required per group ($A$) at a given age and latitude is simply:

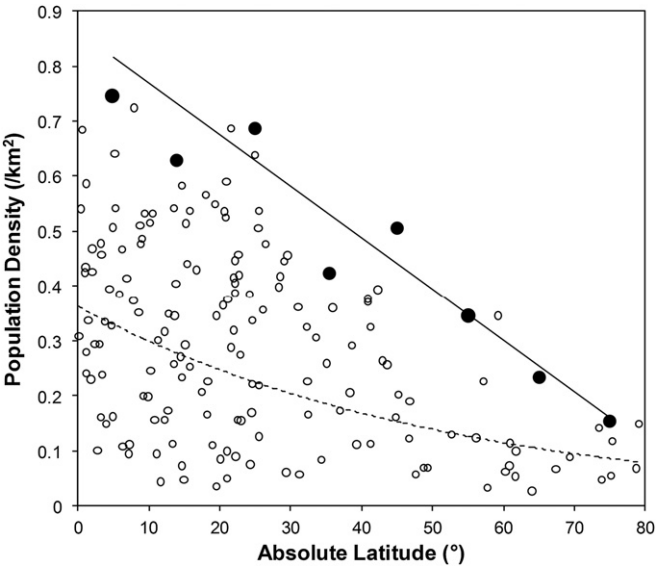



**Figure 1.** Regression of population density on absolute latitude for 181 groups from the Binford (2001) sample. Note that the variance in population density is substantial near the equator, but that a constraint lowering viable population density is clearly visible further from the equator. The larger, filled circles are the binned maxima used for the upper bound regression (straight line). The best fit to the whole dataset is an exponential regression (dashed line). The upper bound line is clearly a more accurate specification of the latitudinal constraint.

$$A_g = \frac{gs}{pd} \quad (6)$$

It should be noted at this point that as Eq. (5) calculates a maximum population density, the areas calculated will necessarily be conservative; groups living at lower population densities would of course require larger areas, and this point will be addressed in the discussion below.

In order to relate the data on the areas required by groups of hominins to the potential fission-fusion dynamics of those groups, each hominin specimen in the dataset is plotted on a bivariate scatterplot with latitude as the abscissa and age as the ordinate. To examine fission points, the area required per group is compared with the area that that group could reasonably be expected to cover during a day's foraging, under the assumption that a cohesive group no longer able to cover its home range in a day could forage more profitably by splitting into subgroups (see Layton and O'Hara, 2010; Layton, in press; and discussion above). The area that a group could cover in a day is calculated using a simple gas model of the kind often applied to primate foraging (e.g., Waser, 1976, 1982; van Schaik and Dunbar, 1990; Mitani et al., 1991; Lowen and Dunbar, 1994; Dunbar, 1995, 2000; Holenweg et al., 1996; Barrett and Lowen, 1998; Harcourt and Greenberg, 2001; Gursky, 2005) and, more recently, to the foraging strategies of human hunter-gatherers (Grove, 2010b). To predict the area covered by a cohesive group during a day's foraging ($A_d$), the gas model equation is of the form:

$$A_d = \frac{n_g 8 v t D}{\pi} \quad (7)$$

Where $n_g$ is the number of groups ($n_g = 1$ when the group is cohesive), $v$ is the group's velocity, $t$ is the time period under consideration, and $D$ is the detection distance of the group. Data collated by Layton (in press) suggest that the maximum daily foraging range of modern hunter-gatherers is between 15 km and 25 km. The current study employs Layton's median figure of 20 km. However, since these measurements are given in terms of the maximum distance that a forager is likely to stray from camp

during the course of a day, and given that foragers do not retrace their steps when returning to camp (as this would involve redundant coverage of recently foraged terrain (see Yellen, 1977; Binford, 1982; Grove, 2009, 2010d)), the appropriate figure for input to the gas model for estimation of area covered is 40 km (20 km out, 20 km back). Since the velocity of a foraging group is highly variable (depending on encounters with resources, etc.), and since Layton's figures relate to foraging range per day, we simply set $vt = 40$, allowing $t$ to equal a single day. Detection distance is rather harder to estimate, but we here follow Dunbar (2009) in estimating the detection distance of a group at $D = 0.5$ km. This accords well with the theoretical work of Vine (1973). A group can cover their home range during a day (and thus forage as a cohesive unit) provided:

$$\frac{n_g 8vtD}{\pi} \geq \frac{gs}{pd} \tag{8}$$

When the inverse of this inequality holds the group should fission, with subgroups moving in different directions so as to maximize the area covered per unit time. We can model this situation by setting $n_g > 1$ to indicate that the group is no longer foraging as a single, cohesive unit. Given the values of $vt$ and $D$ employed here, the area a single foraging group can cover in a day is approximately 51 km$^2$. If the group splits into two ($n_g = 2$), the two new groups can collectively cover approximately 102 km$^2$, and so on with further fission events. Isoclines equivalent to the points at which a group must fission into each subsequent level can be plotted on the age/latitude plot by first regressing group size on age, as follows:

$$\ln(gs) = -0.2282\ \text{age} + 4.5483 \tag{9}$$

($n = 191$, $F_{1,189} = 1098.199$, $R^2 = 0.853$, $p < 0.001$). Subsequently, by substituting Eqs. (5) and (9) into Eq. (6) and re-arranging to solve for age given a particular latitude and the threshold area, $A_d$, at which each fission point $n_g$ occurs, fission point isoclines are given as:

$$\text{age} = \frac{\ln(-0.0104\ lat + 0.8636)\ A_d}{\frac{e^{4.5483}}{-0.2282}} \tag{10}$$

The result of Eq. (10) has associated confidence intervals that are calculated for each individual point by dividing the lower CI of Eq. (9) by the upper CI of Eq. (5) (to get the lower CI of Eq. (10)) and the upper CI of Eq. (9) by the lower CI of Eq. (5) (to get the lower CI of Eq. (10)). These confidence intervals are plotted along with the mean values on the age/latitude graphs to give an accurate impression of the uncertainty involved in the estimates.

## Results

Complete calculations of brain volumes, neocortex ratios, group sizes, population densities, and area requirements for each fossil specimen are given in Table A1. SOM Figs. S1 and S2 compare results obtained via Eqs. (1)–(3) with those obtained directly from Eq. (4).

Fig. 2 shows area required per group for a discrete series of latitude/age combinations, giving an intuitive impression of the relative effects of the two variables. It is clear from this figure that, over the latitudinal and temporal ranges examined in the current paper, latitude has a far greater effect on the area required per group than age. This can be formalized by observing that the proportional difference in area required per group caused by the effects on population density of moving 1° further from the equator is equivalent to that caused by approximately eighty thousand years of group size increase. Regarding the latitudinal gradient

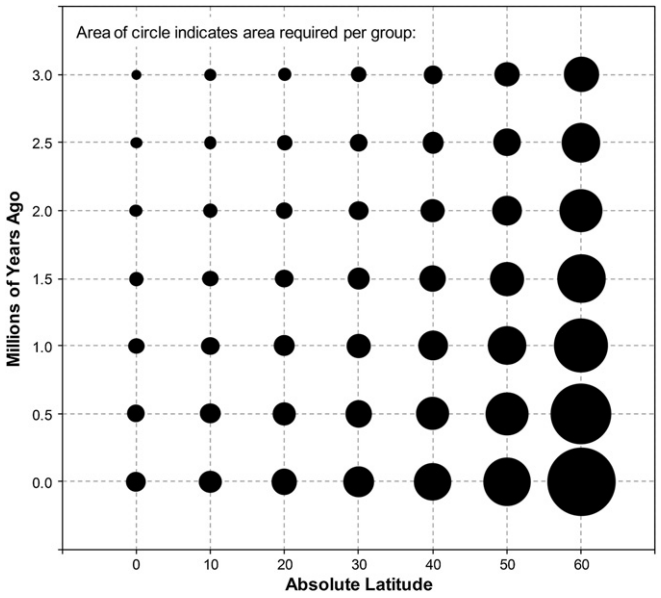



**Figure 2.** The effects of age (as a proxy for group size) and absolute latitude (as a proxy for population density) on the area required per group. An arbitrary area is chosen at 0° latitude, three million years ago. The graph shows how this area would increase, demonstrating that latitude has a far greater effect on areal requirements than age.

alone, the relationship obtained here suggests that an area sustaining a group of 150 at the equator would sustain a group of 113 at 20° from the equator, 77 at 40°, 41 at 60°, and only five at 80°. These observations make explicit the importance of taking latitudinal effects into account when discussing the social ramifications of the hominin diasporas.

Fig. 3 plots the hominin fossil sample by age and latitude against the fission points calculated from Eq. (9). Here the relationship graphed in discrete form in Fig. 2 is converted to a continuous colour scale, with fission points and their associated confidence intervals marked as thresholds. Average areas required for each of the species or groups thereof in our taxonomy are shown in Fig. 4. In this figure, the boxes span the mean estimates of area required, with the whiskers stretching between the lowest and highest confidence intervals within each species. Though any such result is affected by the choice of taxonomy, a number of interesting results arise from this graph and the associated statistics. As none of the samples are normally distributed (a sample for a given species consists of all the mean estimates, upper CLs, and lower CLs for that species pooled), a series of non-parametric Mann–Whitney pairwise tests were undertaken to highlight significant differences between species or groups of species. A number of pairwise tests showed informative and statistically meaningful relationships between the area requirements of different taxa despite the inclusion of individual confidence intervals in the analyses. Firstly, there is a significant difference between the area requirements of the southern and eastern African australopithecine species (Mann–Whitney U, $Z = -2.038$, $p = 0.042$). This result is of course dependent primarily upon the latitudinal difference between eastern and southern Africa, with lower population densities predicted further from the equator. Nonetheless, the result that two contemporary gracile australopiths, such as *A. afarensis* and *Australopithecus africanus*, should have differing area requirements (see Fig. 4) purely due to environmental differences prompted by a latitudinal gradient further reinforces the importance of this variable in scenarios of hominin evolution.

The second significant difference comes in a pairwise comparison of the australopithecines and transitional species (classified

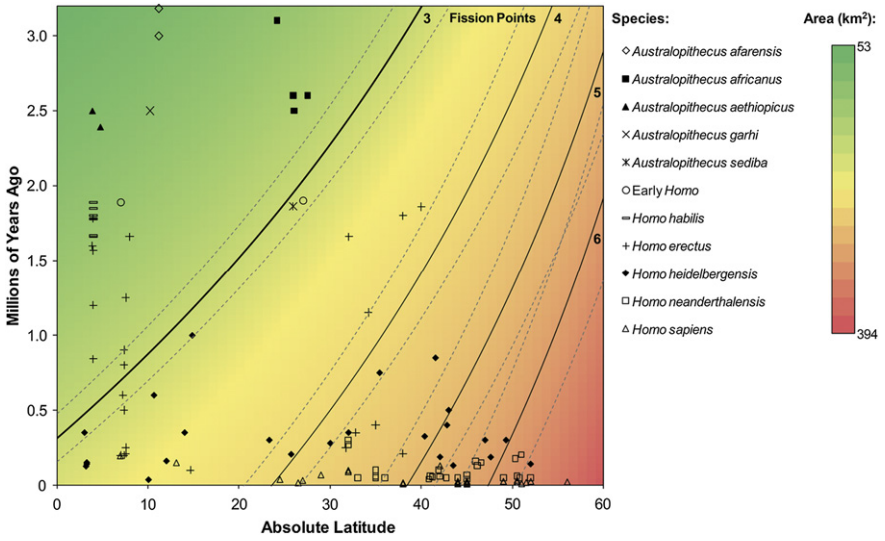



**Figure 3.** Predicted fission points based on Eq. (10). The community is predicted to split into the sub-grouping level indicated to the right of each isocline. For example, *Australopithecus sediba* is estimated to have split its groups into three subgroups for foraging purposes, as it falls to the right of the '3' isocline, but to the left of the '4' isocline. The dashed lines show the confidence intervals associated with each estimated fission point.

here as early *Homo* and *Homo habilis*) as a single group and the first undisputed member of our genus, *Homo erectus* sensu lato (here including some specimens attributed by others to *Homo ergaster*). This difference is highly significant (Mann–Whitney U, $Z = -5.379$, $p < 0.001$), and is a result of the far-reaching occupation of *H. erectus* s.l. in terms of both geography and chronology. A clear difference between *H. erectus* s.l. and the foregoing taxa is apparent in Fig. 3, and indicates a clear break point in hominin strategies, commensurate with the position of *H. erectus* as the first species to disperse throughout Africa and beyond. It should be noted in this context that *H. erectus* persists for at least 1.8 million years (Antón, 2003) and that there is considerable variation in area requirements predicted for this species. Whilst a temporal division seems arbitrary, a division into African and Asian specimens yields

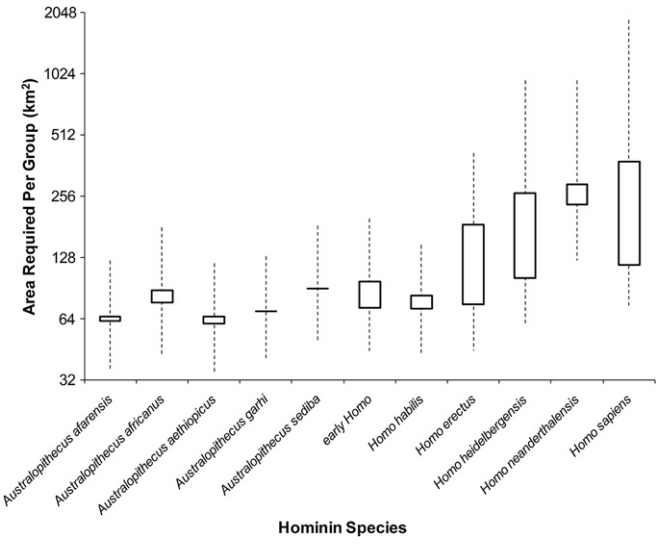



**Figure 4.** Averaged area per group requirements for the lumping taxonomy employed in the current paper, with species displayed according to a broad chronological progression from left to right. The box for each species shows the range of mean area estimates from Table A1, while the whiskers show the distance between the highest upper confidence limit for that species and the lowest lower confidence limit for that species. Associated statistics, where significant, are reported in the text.

a consistent difference (Mann–Whitney U, $Z = -2.013$, $p = 0.017$), suggesting that the areas required by Asian members of this species would have been considerably larger.

A fourth significant difference arising from Fig. 4 and associated statistics is between *H. erectus* and *Homo heidelbergensis* ($Z = -3.426$, $p = 0.001$). Here the larger area requirements of the latter species are due to a combination of larger groups and the expansion of this species into Europe at various times over the last 750,000 years (Gamble, 2009). The final significant difference is between *H. heidelbergensis* and *Homo neanderthalensis* ($Z = -2.820$, $p = 0.006$), the latter being the only species in the sample to occur purely outside Africa (though note that the *H. sapiens* sample here is biased towards later European specimens). In terms of the current analysis, the high latitude adaptation of the Neanderthals (Holliday, 1997; Tzedakis et al., 2007; Hublin and Roebroeks, 2009) relates purely to ranging patterns, which are predicted to have been extensive, though postulated physiological adaptations may have accompanied the maintenance of such vast ranges (Aiello and Wheeler, 2003). It is also of interest that Neanderthal and modern human ranging patterns are not significantly different. This might look rather different, however, if a larger sample of early African *H. sapiens* crania were available or if Neanderthal cranial volumes (and therefore group sizes) were corrected for disproportionately enhanced visual areas (see Pearce and Dunbar, in press).

A number of specific results that emerge from the mapping of Fig. 3 are worth reporting here in further detail. The analysis suggests that all of the hominins in the current sample would have periodically divided their groups into at least two subgroups (the fission point at which this occurs falls outside the region of Fig. 3, to the top left of the graph). Whilst the southern and eastern australopiths would have required significantly different areas to sustain their groups, this does not involve an additional level of fission. Chronologically, the first hominins to cross this third fission line are, bar one, members of our genus. Early *Homo*, *H. erectus*, and the '*Homo*-like' *Australopithecus sediba* (Berger et al., 2010) all cross this boundary shortly after 2 Ma. Whilst the earlier crossings of the third fission line occur in southern Africa and during tentative initial movements into Eurasia (that is, they occur due to the latitudinal constraint on population density), *H. erectus* has crossed the third fission line even in equatorial east Africa by half a million

years ago (due to increases in group size). The persistence of *H. erectus* for over 1.8 Ma and over a total latitudinal span of almost 70° suggests either multiple regional social adaptations (e.g., Rightmire, 1995) or profound cultural versatility (e.g., Grove, 2011a,b). It is notable that *H. heidelbergensis* is the first hominin to exist in chrono-geographic regions exclusively beyond the third fission line, whilst *H. neanderthalensis* exists exclusively beyond the fourth fission line. The diversity of fission structures evidenced in *H. sapiens* (existing beyond lines three-six) is in fact matched by *H. heidelbergensis*, whilst the high latitude focus of the Neanderthals requires them to fission regularly into at least four subgroups. Thus, whilst the medians of *H. sapiens* and *H. neanderthalensis* are not distinguishable statistically, their variances are significantly different (Rank F test, $F_{186,96} = 0.675$, $p = 0.033$), reflecting the wide distribution of *H. sapiens* relative to the Neanderthals.

It should be noted that the model employed above does not take into account the changing climatic regimes of the 3.2 million years surveyed. Though this would not affect comparisons of species living contemporaneously, such as those made above between the southern and eastern australopiths, it could have a bearing on comparisons between species that lived in quite different periods. This is due to the possibility that hominins only moved into higher latitudes during warm phases, and thus were not subject to the constraints imposed on contemporary hunter-gatherers living under the current climate regime. As this issue is of some importance in the interpretation of the model, a preliminary test was run on the relationship between latitude and global climatic fluctuation as it relates to the hominins in the current sample. If there is a tendency in the dataset for higher latitude occurrences to occur during warmer periods, age-specific temperature proxies identified for each specimen should correlate with specimen latitudes. Using the LR04 Stack (Lisiecki and Raymo, 2005), the most comprehensive record of global climatic change available for the period in question, a benthic $\delta^{18}O‰$ value was obtained as an indicator of global temperature at an age equivalent to that of the specimen. As a higher benthic $\delta^{18}O‰$ value indicates a lower temperature, there should be a negative relationship between this value and specimen latitude if higher latitude occurrences tend to occur during warm periods. In fact, the regression is significantly positive ($n = 191$, $F_{1,190} = 9.911$, $R^2 = 0.05$, $p = 0.002$), though it explains less than 5% of the variance in the data. Whilst this result is sufficient to demonstrate that there is not a tendency towards higher latitudes during warm phases alone in this sample as a whole, this issue will require further work as the model is refined, as the scale of the current analyses may miss the effects of specific refugia.

## Discussion

### *Implications of the model*

The model considered here suggests that a trend towards larger group sizes in more recent populations combined with a trend towards lower population densities at higher latitudes led to a dramatic increase in area requirements for hominin groups expanding from the tropics, and that those groups responded to this pressure by developing fission-fusion social systems at varying levels. This scenario is in accord with a number of studies examining patterns of fission and fusion in extant non-human primates (e.g., Chapman, 1990; Chapman et al., 1995; Gillespie and Chapman, 2001), as well as with research examining hierarchically nested grouping structures in modern hunter-gatherer and sedentary groups (Hill and Dunbar, 2003; Zhou et al., 2005; Hamilton et al., 2007). The Ecological Constraints Hypothesis (henceforth ECH) suggests that the area required to meet the energetic and nutritional requirements of a group of animals will increase with group size, resulting in a concomitant increase in travel costs as the group ranges further (Chapman, 1990). Eventually, "some point will be reached at which energy spent in travel exceeds the energy obtained from the environment, and a smaller group size will become advantageous" (Chapman, 1990: 2). This premise has been tested extensively among primates, with the majority of studies reporting either positive correlations between group size, home range size, day range, or percentage time spent feeding (e.g., Gillespie and Chapman, 2001; Ganas and Robbins, 2005; Teichroeb and Sicotte, 2009) or, primarily in fission-fusion societies, negative correlations between resource density and group size (Chapman, 1990; Chapman et al., 1995; Goldsmith, 1999; Chapman and Chapman, 2000). This latter result suggests that fission-fusion allows primate groups to achieve what is effectively a dynamic ideal free distribution (Fretwell and Lucas, 1970; Krivan, 1997; Cosner, 2005), adjusting group sizes at fine temporal scales to variations in resource abundance.

Although the validity of the ecological constraints model has thus far been assessed over relatively short timescales, a number of studies have addressed the evolutionary implications of a scenario in which time, including that spent travelling between resource patches, is a fundamental ecological constraint (Altmann, 1974; Dunbar, 1992b; Korstjens et al., 2006; Korstjens and Dunbar, 2007; Lehmann et al., 2007, 2008). Analyses carried out by Lehmann et al. (2007) suggest that chimpanzees could not survive anywhere in Africa without the reductions in moving time created by their fission-fusion social system. Modelling chimpanzee time budgets without fission-fusion reduces the predicted maximum ecologically tolerable group size to just 16 individuals, smaller than any known chimpanzee community. Korstjens et al. (2006) obtain a similar result whereby modelled spider monkey populations not permitted to fission for foraging would be limited to reproductively non-viable community sizes of five or six individuals. Lehmann et al. (2007) argue that the primary benefit of fission-fusion is that it allows community size to increase due to reductions in travel costs caused by group fission for foraging purposes.

A key component of the model presented here is that reductions in population density with group size held constant will necessitate larger areas per group. However, as pointed out by Chapman et al. (1995), density is one of three components that together characterize habitat structure. The size of individual patches and their distribution in space are variables that can change independently of patch density, and small foraging groups should be favoured primarily when patches are low in density, uniform in distribution, and small in size (Chapman et al., 1995). In all other situations, there is nothing to prevent the community foraging as a cohesive unit. The structure of the habitat is of equal importance to hunter-gatherers, although the scale at which it is perceived may be extended by the emphasis on equitable bonds with neighbouring groups (Grove, 2010b) and the function of those bonds as insurance mechanisms against potential subsistence failure (Whallon, 2006). Whallon (2006) has recently suggested that the most important habitat variable among modern hunter-gatherers may be resource heterogeneity at the inter-community scale, since for the strategy of variance reduction through bonds with neighbouring groups to be effective those groups must inhabit areas that are non-synchronized in their resource potentials. Only when this is true can extended networks avert risks created by unpredictable resource failures.

The research of Korstjens and colleagues (Korstjens et al., 2006; Korstjens and Dunbar, 2007; Lehmann et al., 2007, 2008; Asensio et al., 2009) suggests that, over evolutionary timescales, fission-fusion has arisen to facilitate the formation of larger communities via the elevated foraging efficiency achieved in small subgroups. This logic, combined with the scenario described by the results

presented here, suggests that as community sizes increase, so the size of the foraging group should remain relatively constant, since this smaller group size should be optimized with respect to foraging efficiency in the context of additional constraints such as predation risk and other inherent benefits to grouping. In the hunter-gatherer sample utilised in the current paper, despite population size varying by over two orders of magnitude, and population home range by almost four orders of magnitude, the size of the foraging group is remarkably constant, with 91% of group sizes falling between five and 27 and a median of 16, mean of 17, and mode of 18. This suggests that hunter-gatherers requiring larger areas to support their populations due to constraints on population density or other factors manage those areas by fissioning into ever-greater numbers of lower-level groups. That is, the size of the lower-level grouping remains approximately constant, but the number of groups of this size increases in linear fashion as the overall size of the area utilized increases. Regressing total area utilized against the ratio of foraging group size (Binford's (2001) 'Group 1') to total population size (an index of the number of foraging groups that the population divides into), demonstrates that this is indeed the case ($F_{1,187} = 94.57$, $R^2 = 0.336$, $p < 0.001$; see Fig. S3). This suggests that the function of fission-fusion grouping patterns in humans may be the same as that among non-human primates and other animals, even though the former occurs over much larger spatio-temporal scales.

Another postulated difference between human fission-fusion social systems and those of other primates is the presence of a number of hierarchical layers rather than the simple division of a community into foraging parties found in most primate fission-fusion systems. In this sense human society is 'multi-level' rather than strictly fission-fusion as understood in primatological terms (Kappeler and van Schaik, 2002), but again this classification is inadequate to fully describe hunter-gatherer society. The multi-level groups of hunter-gatherers are far more fluid in terms of composition than those of typical multi-level primates such as hamadryas baboons, and this fluidity is often regarded as a hallmark of hunter-gatherer society (Yellen and Harpending, 1972; Peterson, 1975; Yellen, 1977; Howell, 1979; Littleton and Allen, 2007). In this respect, a human society is far more like a chimpanzee society that has added additional levels than any of the multi-level societies currently documented by primatologists. This raises the questions of when, why, and how these additional levels were added. The question of how they were added essentially asks whether levels were added below (at the subgroup level, by fission) or above (at the super-group level, by fusion). The analyses presented here and culminating in Fig. 4 are an attempt to answer the when and the why. The how is rather more contentious, but the foregoing analyses do provide a tentative answer. The analysis of Group 1 can be extended to demonstrate that not only the smallest level of grouping in the hunter-gatherer sample, but also Binford's Group 2 and Group 3 aggregations, when taken as ratios of population size, correlate with area required per group (Group 2, $F_{1,177} = 57.09$, $R^2 = 0.244$, $p < 0.001$, see Fig. S4; Group 3, $F_{1,151} = 38.51$, $R^2 = 0.203$, $p < 0.001$, see Fig. S5). Group 2 is defined as the mean size of the largest residential seasonal camps, whilst Group 3 is mean size of periodic regional camps (Binford, 2001). The fact that all three of Binford's listed social groupings vary in this way provides a provisional explanation for the way in which relations between levels are managed. Hunter-gatherers adjust to increasing group sizes not by scaling up the sizes of their lower-level groupings accordingly, but by increasing the numbers of groups in each lower-level. Thus if a population of 135, with an average Group 1 size of five, Group 2 size of 15, and Group 3 size of 45, were to grow over time to a population of 270 individuals, it would not double the size of its subsidiary groups to ten, 30, and 90, but simply maintain twice as many groups at the original sizes of five, 15, and 45.

### *Caveats and potential extensions*

The foregoing framework provides a series of inferences about evolving hominin social systems. However, there are necessarily a series of factors not considered in the current paper that could potentially provide further insight into the issues raised. These can be broadly characterised as relating to changes in body mass, diet, technological efficiency, and locomotor adaptations, and each is discussed in turn below.

Over the three million years considered by our hominin sample and the related modelling, hominin body size has increased, with the most notable change occurring with the appearance of the earliest indisputable members of our genus, as represented by *H. ergaster* or *H. erectus*. As established at length by Damuth (1981a,b, 1987), heavier animals require more resources and thus live at lower population densities than lighter animals. If this factor were included in the current analyses, it would strengthen the temporal trend towards higher area requirements in more recent hominins. The conclusions presented here can thus be viewed as being conservative with regard to this trend. There is also a more subtle effect of body mass as it relates directly to fission-fusion social systems. Lehmann et al. (2007; Lehmann and Dunbar, 2009) note that fission-fusion as a solution to foraging competition may only be open to relatively large-bodied animals near the top of the food chain. Predation risk is one of the primary pressures considered to favour the formation of large groups. Animals that are subject to this pressure would be taking extreme risks by fissioning into smaller parties to forage. However, primates of large body size are considerably less likely to fall victim to predators and this may afford them greater freedom to forage in subgroups of varying size (see Dunbar et al., 2009; Bettridge et al., 2010). In terms of human evolution, again the distinction appears to fall between the smaller bodied australopithecines and the larger members of the genus *Homo*, with the latter arguably operating at a higher trophic level (see e.g., Brain, 1981). It is of interest given this scenario that both southern and eastern australopithecines retained arboreal adaptations that might have enabled them to avoid terrestrial predators. This is a potential solution to the problem of elevated predation risk in smaller groups, but would have restricted these animals to foraging in the vicinity of suitable vegetation.

Further differences between hominin species over the time period in question relate to diet, technological accomplishment, and locomotor dynamics. Evidence from dental, maxillo-facial, and cranial morphology (e.g., Ungar, 1998, 2004; Teaford and Ungar, 2000; Teaford et al., 2002; Daegling and Grine, 2007), as well as stable isotope and trace element analyses of fossil bone (e.g., Sillen et al., 1995; Lee-Thorp, 2002; Sponheimer et al., 2007) and the evidence of evolving gut morphology (e.g., Aiello and Wheeler, 1995; Milton, 1999, 2003) suggest an increasing reliance on meat during the course of human evolution. Carnivores require substantially larger home ranges than primary consumers (Milton and May, 1976; Harestad and Bunnell, 1979; Leonard and Robertson, 2000; Antón et al., 2002), while a related issue is that that the type of prey hunted may impact directly on optimal group size (see e.g., Caraco and Wolf, 1975; Pulliam and Caraco, 1984; Giraldeau and Gillis, 1988; Creel and Creel, 1995). The first stone tools may have been crucial in enabling hominins to butcher carcasses (e.g., Bunn, 1981; Blumenschine and Selvaggio, 1988; Kimbel et al., 1996; Roche et al., 1999; Blumenschine and Pobiner, 2007; Shea, 2007), with cutting edges indispensable in processing soft tissue and percussion tools essential for marrow extraction (Bunn, 2007; McPherron et al., 2010). It is clear that chimpanzee

toolkits provide access to otherwise unobtainable sources of nutrition (e.g., Suzuki et al., 1995; Bermejo and Illera, 1999; Carvalho et al., 2008; Sanz et al., 2009), and that hominin toolkits have been growing in complexity through time (Isaac, 1972; Gowlett, 2009). Finally, although all the species in our sample are obligate bipeds, there are frequent arguments to the effect that australopithecine bipedalism was less efficient than that of our genus (e.g., Steudel, 1996; Wang et al., 2003, 2004; Pontzer et al., 2009; though see Raichlen et al., 2010). In particular, the shorter limb length of australopithecines may have forced them to transition from walking to running at lower speeds (Kramer and Eck, 2000), whilst the postulated 'bent-hip, bent-knee' gait (Stern and Susman, 1983) would have required more muscle power per unit distance (Wang et al., 2004), ultimately limiting these species to smaller day and home ranges. If factored in, each of these broad temporal trends – towards higher levels of carnivory, more sophisticated lithic technologies, and increased bipedal locomotor efficiency – would only serve to strengthen the general pattern of the results presented here.

## Conclusions

A trend towards increasing group size combined with a pattern of repeated expansion from tropical towards temperate latitudes led to a dramatic increase in the area requirements of hominin groups over the past three million years. Larger, more recent groups forced to live at lower population densities in higher latitude regions would have been forced to adjust their foraging patterns so as to avoid unsustainable increases in foraging time. The analyses presented in the current paper, when combined with evidence drawn from primatology and hunter-gatherer ethnography, suggest that the development of increasingly complex, multi-level fission-fusion social systems could have freed hominins of this constraint. Analyses of the fossil record suggest latitudinally-driven differences in the area requirements of 1) australopithecines from East and South Africa, and 2) African and Asian *H. erectus*, whilst chronologically-driven differences appear between 3) *H. erectus* as a whole and *H. heidelbergensis*, and 4) between *H. heidelbergensis* and the Neanderthals. The results suggest a model of human social evolution whereby new grouping levels of an ultimately multi-level society are added as the existing maximum group size hits each of a series of critical fission points.

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## Appendix. Supplementary Data

Supplementary data related to this article can be found online at doi:10.1016/j.jhevol.2011.10.012.

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